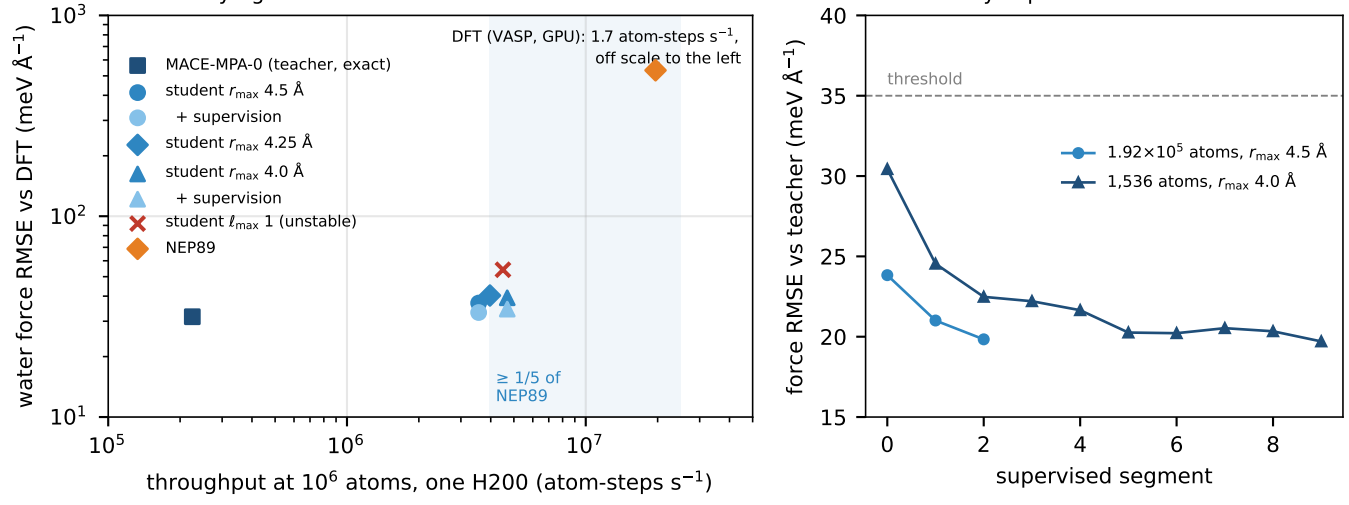
a accuracy against cost
water force RMSE vs DFT (meV Å^{-1})
throughput at 10^6 atoms, one H200 (atom-steps s^{-1})
MACE-MPA-0 (teacher, exact)
student r_{max} 4.5 Å
+ supervision
student r_{max} 4.25 Å
student r_{max} 4.0 Å
+ supervision
student ℓ_{max} 1 (unstable)
NEP89
DFT (VASP, GPU): 1.7 atom-steps s^{-1}, off scale to the left
≥ 1/5 of NEP89
b on-the-fly supervision
force RMSE vs teacher (meV Å^{-1})
supervised segment
threshold
1.92×10^5 atoms, r_{max} 4.5 Å
1,536 atoms, r_{max} 4.0 Å